Learning from Guidelines: Structured Prompt Optimization for Expert Annotation Tasks

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Motivations & Challenges:

- Existing models require massive annotated data to train a neural network which can be labor-intensive.
- Large Language Models (LLMs) can directly conduct general tasks by different prompts. But for more complex applications, they need to follow specific *guideline* to complete the task step-by-step.

Human-Style Learning:

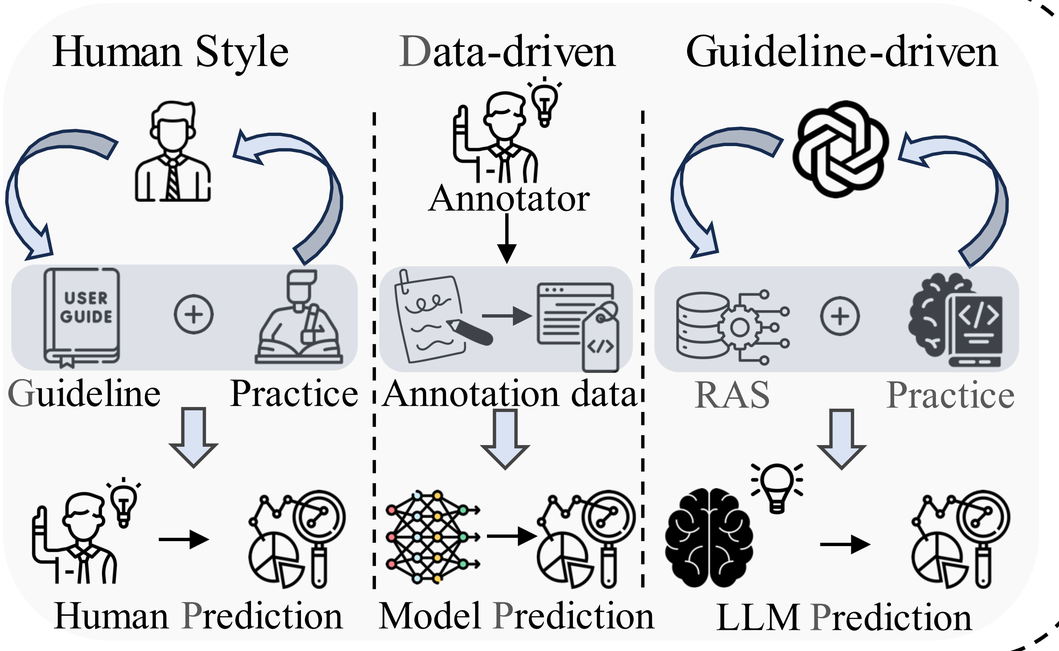
A student mainly *learns from a textbook* and consolidates the understanding through few practice.

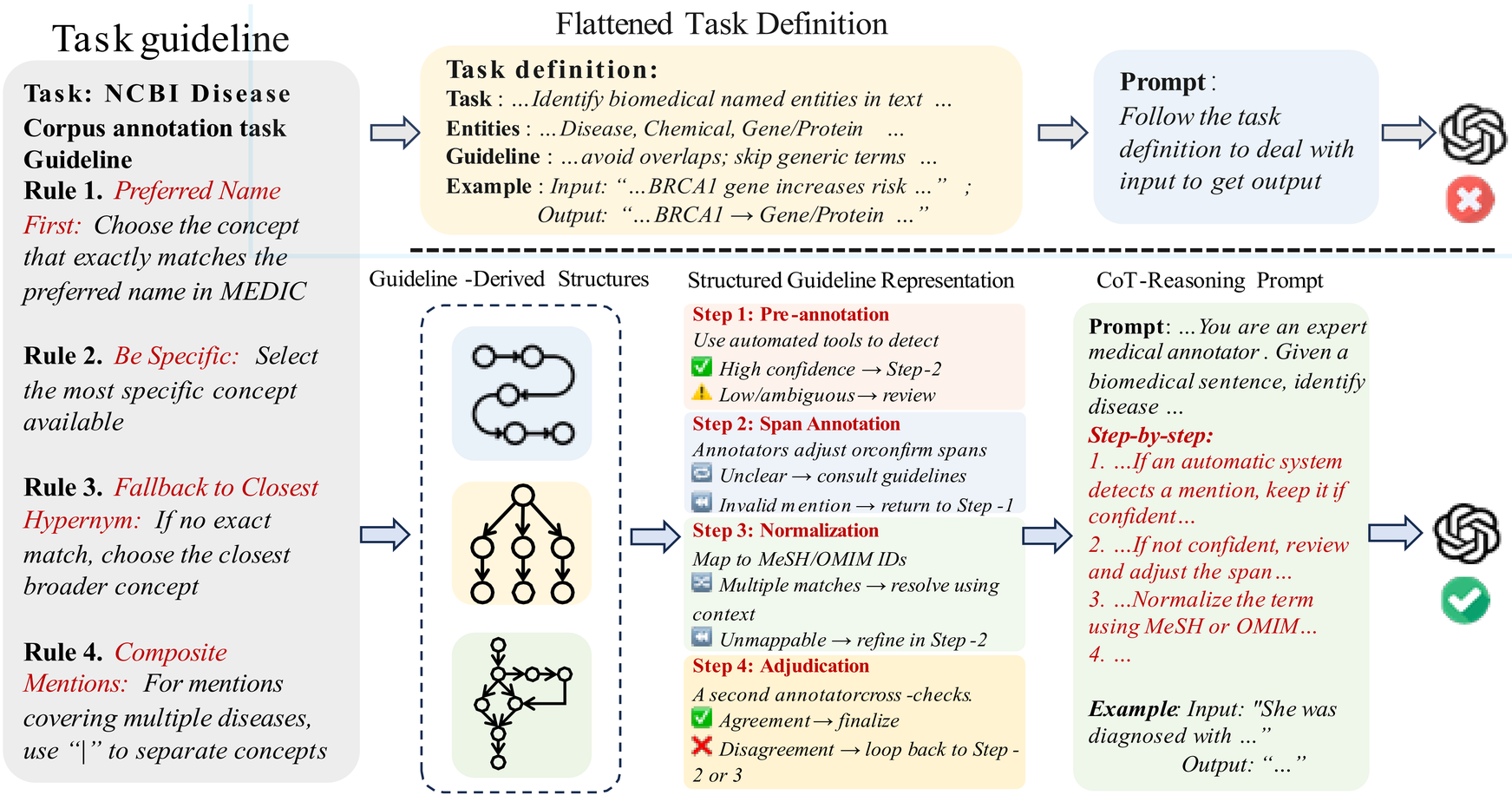
Data-Driven Learning:

Modern neural networks usually rely on massive annotated samples for training to directly *learn from data*.

Guideline-Driven Learning:

Can an LLM *learn mainly from a guideline* like human, and optimize with just few samples?





Guideline Structure

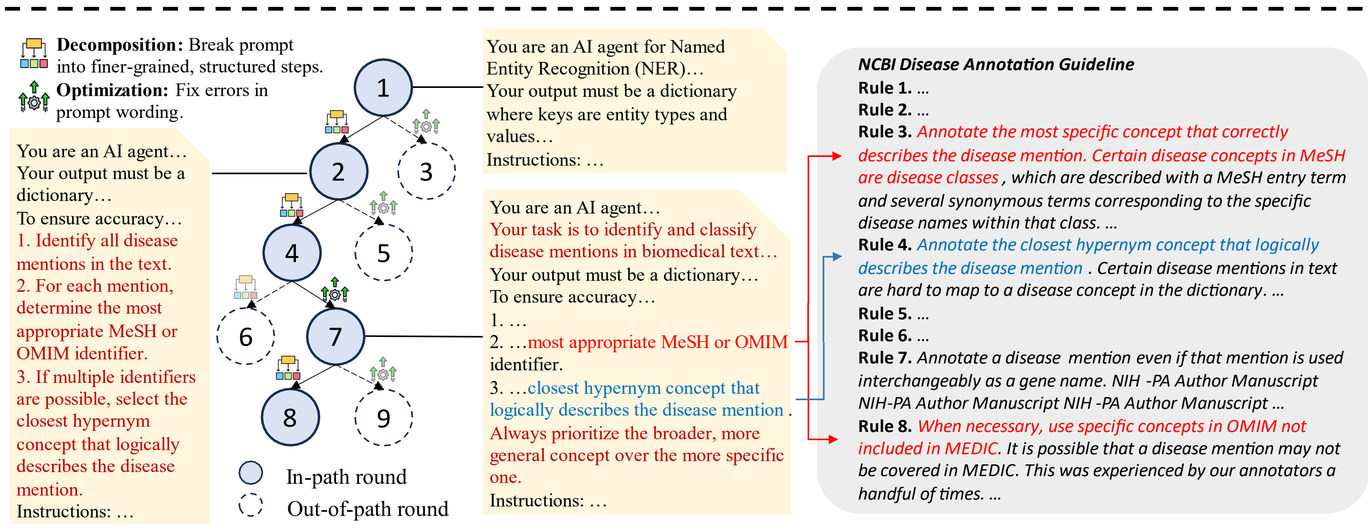
- Real-world guidelines have various formats based on the domain-specific needs.
- We identify most guidelines can be represented as three abstract structures.

Prompt Representation

- Though guideline structures could be different, the exact execution process is step-by-step. It can be presented as a CoT prompt.

Guideline-driven Prompt Optimization: We propose a tree-search algorithm to find the optimal CoT prompt to represent the guideline by few-shot examples. We use a guideline-driven optimizer based on the Retrieval Augmented System. The algorithm starts with a simple task-definition prompt and iteratively derives new prompts after practicing on examples. The optimization is conducted on both the content and structure perspectives after receiving feedback from the guideline. Details can be found in the paper.

Baselines	ICL	GPT-3.5-Turbo					GPT-4				
		NCBI	WNUT	Harv.	CASIE	Avg	NCBI	WNUT	Harv.	CASIE	Avg
Standard Prompting†	1-shot	33.3	41.4	27.2	69.6	42.9	39.2	47.6	39.1	76.2	50.5
	5-shot	45.0	39.6	26.1	74.8	46.4	56.8	50.1	35.6	80.6	55.8
RAS Prompting†	1-shot	35.5	40.4	25.6	70.0	42.9	39.9	48.4	41.0	78.3	51.9
	5-shot	46.5	36.6	17.8	73.1	43.5	60.3	44.9	41.8	81.3	57.1
CoT Prompting†	1-shot	33.8	45.6	33.0	71.2	45.9	31.3	52.6	40.0	77.7	50.4
	5-shot	51.7	43.7	38.1	75.5	52.3	58.5	49.0	43.7	82.5	58.4
AnnoLLM† (NAACL'24)	1-shot	37.4	44.6	16.5	68.3	41.7	45.5	50.2	36.5	74.1	51.6
	5-shot	40.9	43.5	23.9	74.7	45.8	52.1	51.4	40.8	77.7	55.5
PromptAgent (ICLR'24)	1-shot	25.4	45.7	34.1	70.6	44.0	39.3	54.3	35.0	78.2	51.7
	5-shot	42.3	44.2	28.8	72.5	47.0	57.0	49.6	38.7	79.9	56.3
GPO (AAAI'25)	1-shot	26.2	43.0	9.6	68.7	36.9	36.0	48.5	18.7	77.4	45.2
	5-shot	38.7	37.8	15.9	74.8	41.8	55.6	49.5	26.9	80.2	53.1
GDP (ours)	1-shot	**40.1**	**48.4**	**38.3**	**75.0**	**50.5**	**48.5**	**55.5**	**46.8**	**81.6**	**58.1**
	5-shot	**53.5**	**47.1**	**44.7**	**78.8**	**56.0**	**62.1**	**56.5**	**49.5**	**83.4**	**62.9**



Tasks & Datasets: We conduct experiments on domain-specific information extraction tasks covering various professional domains. We use the guideline and few-shot examples to optimize the prompt, which is different from existing prompt optimizations. The optimized prompt can be combined with ICL.

Experiments: We compare with several baselines from different aspects. Experiments demonstrate the effectiveness of our method. The optimized prompt successfully encodes structured guideline knowledge.